

Finger-Vein Biometric Recognition: A Comparative Study of ResNet-18 and VGG-16

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Abstract

- › About the Project
- › Dataset
- › Image Pre-Processing
- › Data Preparation
- › Model, Training and Score Extraction
- › Evaluation
- › Conclusions



About the project ●○

Reference Paper

Finger-Vein Biometric Recognition: A Review

Borui Hou[✉], Huijie Zhang[✉], and Ruqiang Yan[✉], *Fellow, IEEE*



About the project ○●

Introduction

- This project presents the design and evaluation of a **finger vein-based** biometric recognition system using deep learning models such as **ResNet-18** and **VGG-16**
- The vein patterns are unique, permanent, difficult to replicate
- The experiments for this project were conducted using the **MMCBNU 6000** dataset ^[4]



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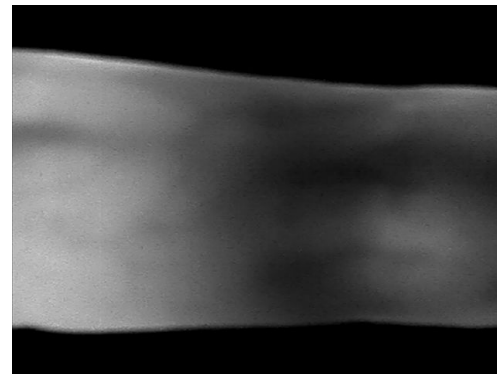
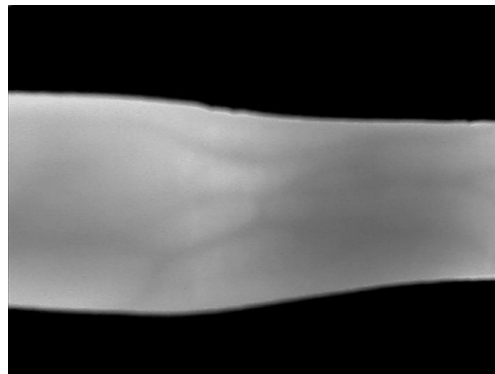
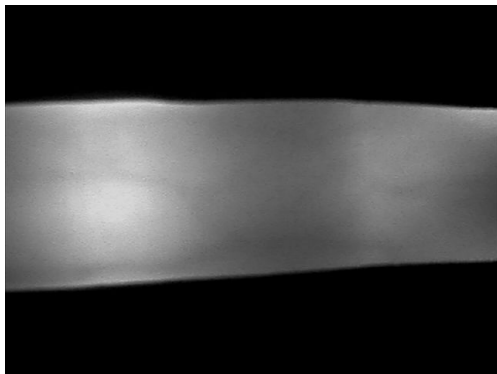
- **MMCBNU 6000** was designed to provide a large-scale, demographically diverse benchmark for fingervein recognition research
- It was collected from **100 subjects** (83 males and 17 females) originating from 20 different countries, with ages ranging from 16 to 72 years.
- For each subject, six fingers were imaged.
- For each of these six fingers, ten samples were acquired ($100 \times 6 \times 10 = \mathbf{6000}$)



Dataset ○●

Dataset

- MMCBNU 6000 images were captured in a **conventional office setting**, with variability in illumination and imaging conditions





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Image Pre-Processing

Raw images suffer from **position and orientation variations, uneven contrast, and background noise**, affecting feature extraction and matching.

Pre-processing Pipeline:

1. Image Boundary Detection
2. Finger Alignment
3. Contrast Enhancement
4. ROI Extraction & Normalization

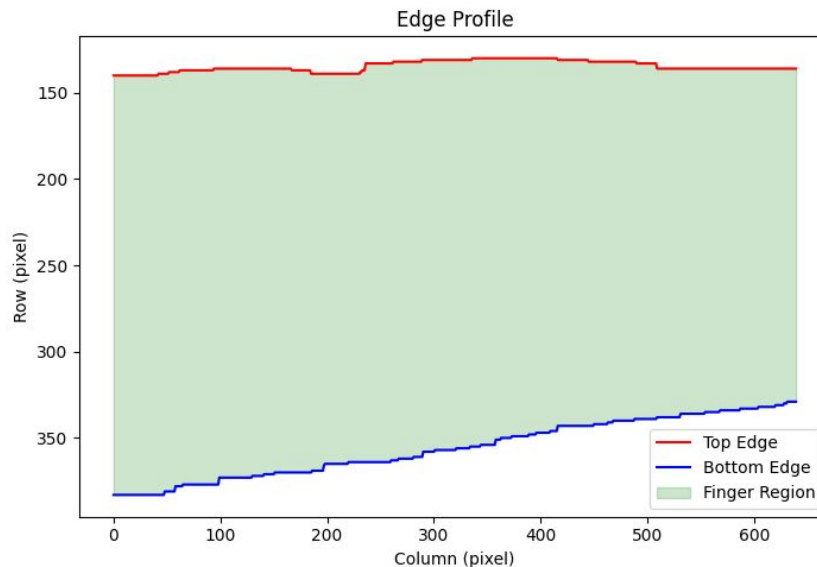
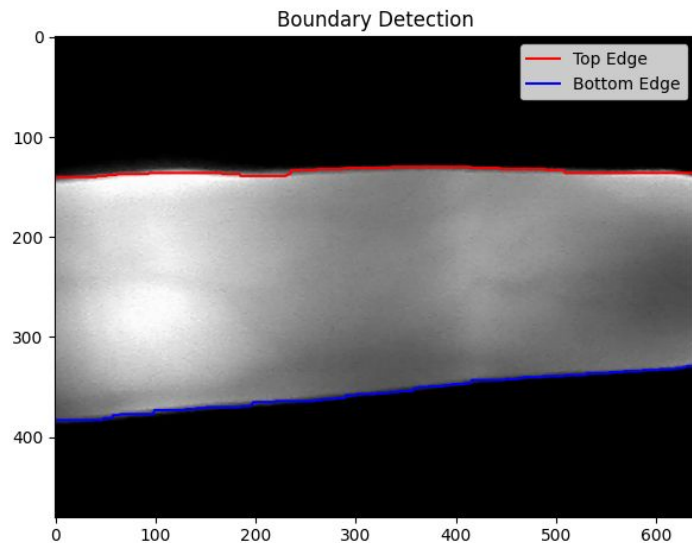


Image Boundary Detection

- **Directional edge-response filtering** based on Y. Lu^[2]
- Image **vertically padded** to avoid convolution border artifacts.
- Two complementary **4×20 gradient masks** detect upper and lower finger edges.
- Identify **boundary candidates** from the maximum filter responses.
- Apply **median filtering** to smooth the detected boundaries.
- Perform a **consistency check** to remove local artifacts and outliers.
- Obtain the final **smooth upper and lower boundary profiles**.



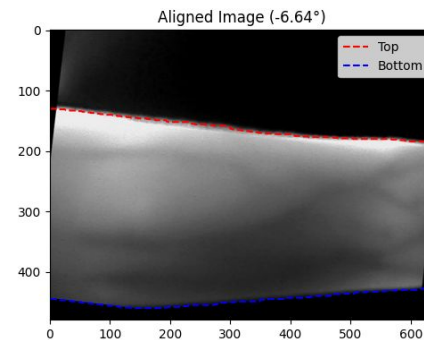
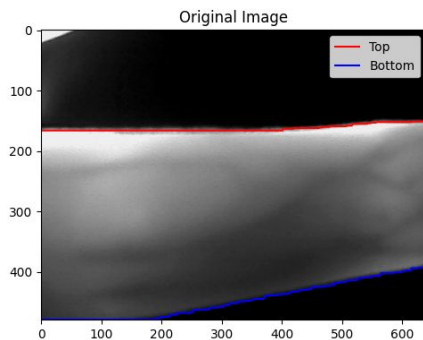
Image Boundary Detection





Finger Alignment

- Compute the **finger centerline** from the upper and lower boundaries.
- Estimate the **finger orientation** using linear regression.
- **Rotate the image** to horizontally align the finger axis.
- Re-detect the **finger boundaries** after rotation for accurate alignment.





ROI Extraction and Normalization

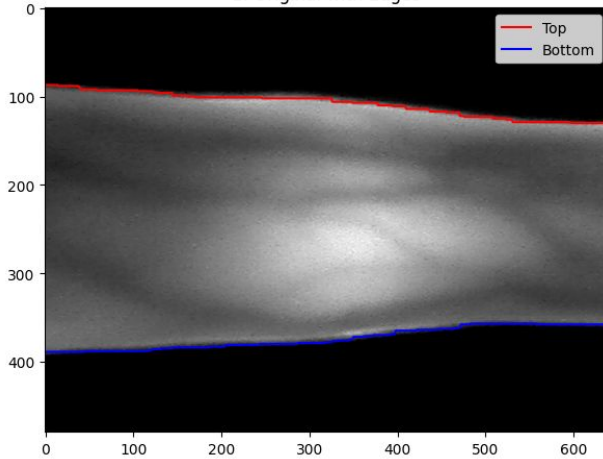
- **Strip-based normalization** inspired by Yang et al.^[3]
- Crop the image to the **central finger region**, removing tip and base.
- Extract a **vertical strip centered on the local finger centerline**.
- Remove **invalid/background columns** based on intensity criteria.
- **Normalize each valid column** to compensate for finger-width variations.
- Fill discarded columns using the **nearest valid column**.
- Resize the final strip to a **fixed ROI resolution**.



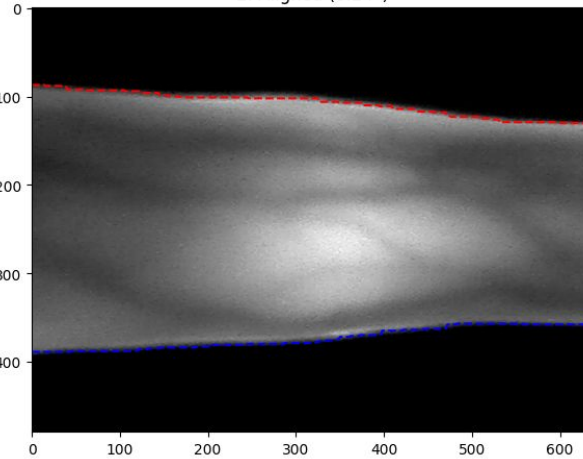
Image Pre-Processing ○○○○○●○

ROI Extraction and Normalization

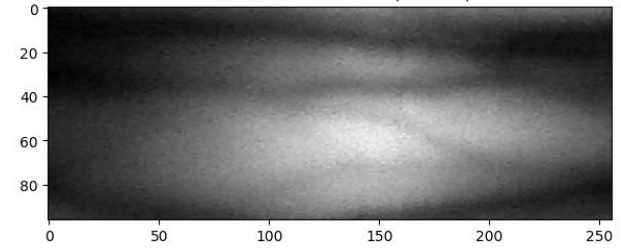
1. Original with Edges



2. Aligned (0.24°)



3. Final ROI Normalized (256x96)





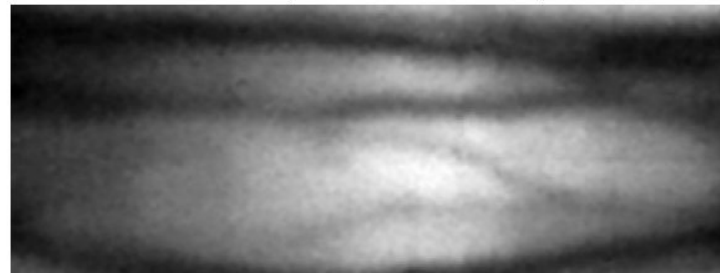
Contrast Enhancement

- Apply **Gaussian blur** to reduce sensor noise.
- Perform **min–max normalization** to rescale image intensities.
- Apply **CLAHE** for local contrast enhancement.
- **Enhance vein visibility** while limiting noise amplification.
- Store the **normalized enhanced ROIs** for feature extraction and classification.

Original ROI



Final ROI (CLAHE + Normalization)





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Two-Fold Cross Validation

- Dataset is divided into **two subject-disjoint folds**
- In each fold:
 - **50% of subjects** → **Training**
 - **50% of subjects** → **Testing**
- Subjects in the test set are **never seen during training**
- Each fold is used once for **training** and once for **testing**



Custom Dataset

- Custom **FingerVeinDataset** implemented using PyTorch
- Each finger is assigned a **unique class label**
- Labels are used to identify the **different finger identities**
- At test time, unseen fingers are represented as **embeddings**
- **300 training identities + 300 unseen test identities** per fold



Data Transformation Pipelines

Training

- Resize ROI to **224×224**
- Apply **data augmentation**: random rotation and color jittering
- Convert images to **tensors**
- Normalize using **ImageNet statistics**

Testing

- Resize ROI to **224×224**
- Convert to **tensors**
- Apply **ImageNet normalization**
- **No augmentation** for deterministic evaluation



Pooling Strategy

- Each fold generates **genuine and impostor similarity scores**
- Scores from **Fold 1 and Fold 2** are concatenated
- Genuine and impostor scores are **pooled separately**
- Pooling is performed **independently for each backbone**
- All **100 subjects** contribute exactly once



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Architecture and Training

- **Backbones & Projection: ResNet-18** and **VGG-16** (pretrained on ImageNet) adapted with a custom 512-D embedding head.
- **Loss Function: ArcFace Loss** (300 identification classes) enforcing an additive angular margin m on the unit sphere for tight intra-class compactness and maximum inter-class separation.
- **Optimization:** Adam optimizer ($lr = 10^{-4}$, **weight decay** = 10^{-4}) with a Cosine Annealing learning rate scheduler trained over 20 epochs.



Score Extraction

- Extract **512-D embeddings** and apply **L2 normalization**
- Compute **pairwise cosine similarity** between all test embeddings
- Consider only the **upper triangular pairs** to avoid duplicates and self-comparisons
- **Genuine pairs**: same finger identity
- **Impostor pairs**: different finger identities
- The resulting **similarity scores** are used to compute ROC, DET, AUC and EER



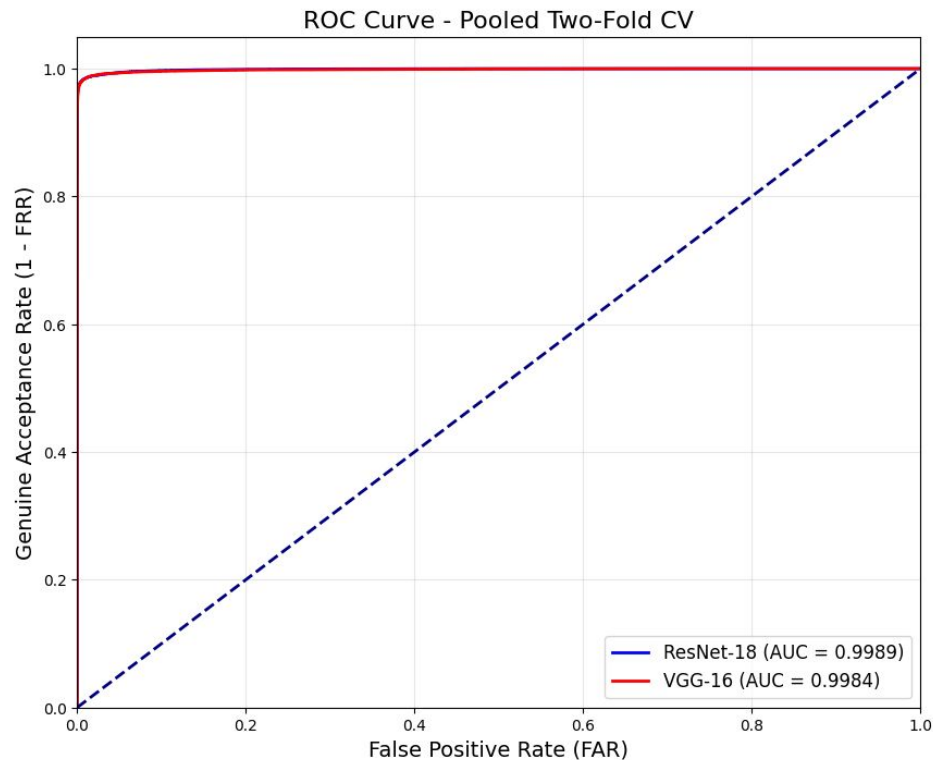
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Evaluation ●○○○

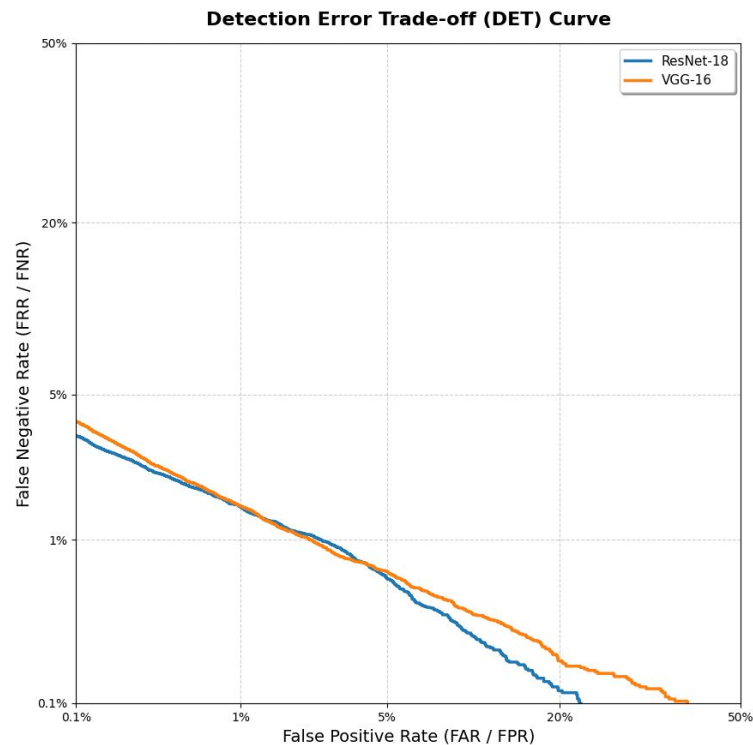
Receiver Operating Characteristic (ROC)





Evaluation ○●○○

Detection Error Trade-Off (DET)

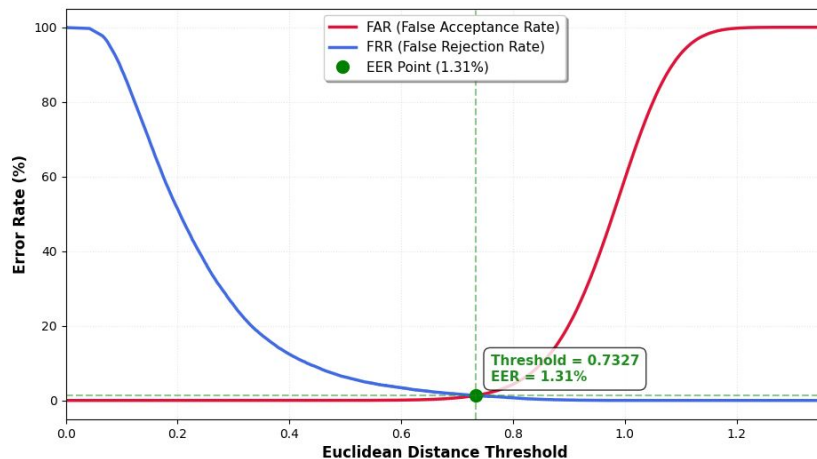




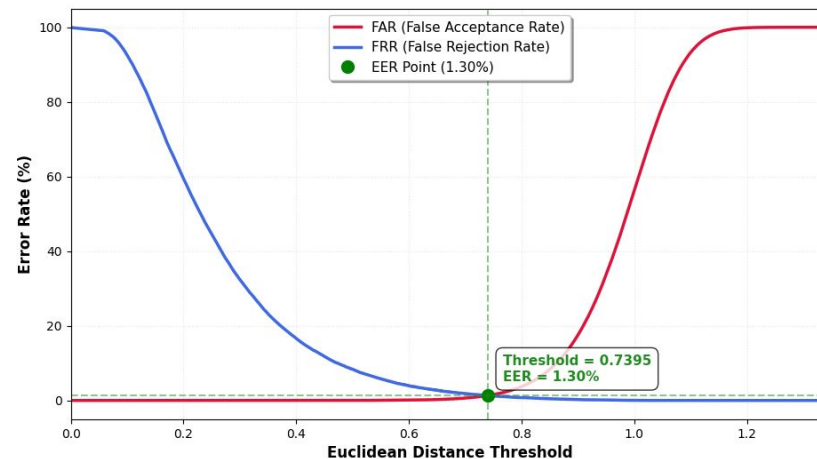
Evaluation ○○○○

Equal Error Rate (EER)

EER (FAR vs FRR) - ResNet-18 (Pooled Scores)



EER (FAR vs FRR) - VGG-16 (Pooled Scores)





Evaluation ○○○●

Efficiency Metrics

Model	Parameters (M)	Inference Time (ms/sample)
ResNet-18	11.44	1.10
VGG-16	14.98	4.95

- **ResNet-18**: fewer parameters and significantly faster inference
- Similar verification performance with **lower computational cost**
- **Best performance-efficiency trade-off**: ResNet-18



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Conclusions •

Comparison with Other Methods

Method	Database	EER(%)
Cond. GAN + CNN	HKPU	1.814
Cond. GAN + CNN	SDU	3.934
CNN recovery	HKPU	2.70
CNN recovery	USM	1.42
CycleGAN	THU-FVDT2	1.12
<u>Inter-class Aug. + Cosine/Triplet</u>	<u>MMCBNU_6000</u>	<u>0.21</u>
Composite + CNN	HKPU	0.33

Method	Database	EER(%)
Composite + CNN	SDU	2.35
<u>Siamese network</u>	<u>MMCBNU_6000</u>	<u>0.17</u>
Pretrained VGG-16	Proprietary	0.396
Pretrained VGG-16	Proprietary	1.275
Pretrained VGG-16	SDU	3.906
<u>Pretrained VGG-16</u>	<u>MMCBNU_6000</u>	<u>0.30</u>
ResNet-18 (this work)	MMCBNU_6000	1.305
VGG-16 (this work)	MMCBNU_6000	1.304



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Thank you for the attention!



References

- [1] B. Hou, H. Zhang, and R. Yan, “Finger-vein biometric recognition: A review,” *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–26, 01 2022.
- [2] Y. Lu, S. J. Xie, S. Yoon, J. Yang, and D. S. Park, “Robust finger vein roi localization based on flexible segmentation,” *Sensors*, vol. 13, no. 11, pp. 14 339–14 366, 2013. [Online]. Available: <https://www.mdpi.com/1424-8220/13/11/14339>
- [3] J. Yang, J. Wei, and Y. Shi, “Accurate roi localization and hierarchical hyper-sphere model for finger-vein recognition,” *Neurocomputing*, vol. 328, 08 2018.
- [4] S. Y. Z. W. D. S. P. Yu Lu, Shanjuan Xie, “An available database for the research of finger vein recognition.”